

HEBRON, NE, USA

WMO: 722124

Lat: 40.149N

Lon: 97.587W

Elev: 1473

StdP: 13.93

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04998

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.7	5.2	-8.2	3.7	3.4	-3.6	4.8	8.1	27.0	29.8	24.6	31.8	6.2	340	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.4	96.7	73.3	93.4	73.0	90.6	72.3	77.8	89.4	76.2	87.6	74.8	85.9	11.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.3	135.6	84.6	72.8	128.4	82.7	71.7	123.8	81.5	42.3	89.7	40.7	87.9	39.2	85.8	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.6	21.0	18.9	DB	-8.1	101.7	5.0	3.0	-11.7	103.9	-14.6	105.7	-17.4	107.3	-21.0	109.5	(4)
			WB	-8.7	80.2	4.8	1.6	-12.2	81.4	-15.0	82.3	-17.6	83.2	-21.1	84.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.3	27.8	30.1	43.2	52.6	63.6	74.5	77.7	74.6	67.9	54.5	41.8	29.7
	DBStd	20.06	11.27	11.98	11.25	9.61	8.82	6.40	5.95	5.77	8.26	9.46	9.84	10.61
	HDD50	2577	688	563	269	82	7	0	0	0	1	60	278	629
	HDD65	5534	1153	978	677	382	137	6	1	4	63	345	695	1093
	CDD50	3780	0	5	60	161	428	735	857	762	539	200	32	1
	CDD65	1261	0	0	2	11	93	290	394	300	151	20	0	0
	CDH74	13446	0	1	51	182	1052	3106	4358	2888	1549	249	10	0
	CDH80	5757	0	0	8	42	385	1378	2085	1202	596	60	1	0
Wind	WSAvg	8.0	7.7	8.4	9.7	10.3	8.8	8.1	6.3	5.9	7.3	7.7	8.0	7.4
Precipitation	PrecAvg	28.0	0.7	0.8	2.0	2.5	4.2	4.0	3.7	3.1	3.0	1.9	1.3	0.7
	PrecMax	45.3	1.9	2.6	8.0	6.7	9.2	10.7	15.0	10.1	9.4	6.9	4.7	2.4
	PrecMin	13.8	0.0	0.0	0.0	0.1	0.8	1.1	0.1	0.2	0.2	0.0	0.0	0.0
	PrecStd	6.4	0.5	0.6	1.7	1.4	2.4	2.2	2.8	1.9	2.1	1.7	1.1	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.3	69.7	79.3	85.7	94.4	99.2	100.0	98.8	96.0	87.9	74.9	60.7
		MCWB	44.4	49.7	58.4	61.9	65.9	70.8	73.6	74.0	70.4	67.3	56.3	47.7
	2%	DB	53.9	61.0	73.3	79.1	88.3	95.0	96.9	93.8	90.7	80.6	68.1	54.7
		MCWB	42.9	47.5	55.7	60.5	66.4	72.9	74.6	73.8	70.5	62.7	53.7	45.1
	5%	DB	48.2	54.2	68.2	74.5	84.1	91.4	93.6	90.3	87.2	76.4	63.1	50.0
		MCWB	40.2	43.8	54.4	58.3	65.6	71.8	74.5	73.7	69.6	60.4	51.1	42.2
	10%	DB	43.1	47.1	62.4	70.0	80.5	88.1	90.4	86.6	82.4	71.8	58.4	45.2
		MCWB	37.1	39.9	51.4	56.3	64.0	70.7	73.9	72.7	67.1	58.0	48.8	39.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.4	54.5	64.1	65.6	73.2	78.3	80.4	79.5	75.1	70.8	59.9	52.6
		MCDB	58.4	61.5	74.0	75.7	85.0	90.5	92.4	90.1	86.7	81.2	65.7	57.1
	2%	WB	43.8	48.5	60.4	63.3	70.5	76.0	78.3	77.3	73.3	67.6	56.3	45.9
		MCDB	52.7	59.9	67.8	74.6	81.3	88.0	90.0	88.4	85.2	77.0	65.7	52.5
	5%	WB	40.5	44.3	56.3	60.8	68.5	74.2	76.8	75.5	71.7	63.0	52.6	42.3
		MCDB	47.6	52.9	65.0	71.1	79.6	86.5	88.4	86.0	82.6	71.6	61.2	49.8
	10%	WB	37.6	40.5	51.8	58.1	66.3	72.6	75.2	74.0	69.9	59.8	49.2	39.1
		MCDB	42.9	47.8	62.9	67.7	77.0	84.6	86.6	83.9	79.2	68.4	58.1	44.7
Mean Daily Temperature Range	5% DB	MDBR	19.6	20.2	23.3	23.6	22.2	22.0	21.4	21.3	24.1	24.3	23.0	19.2
		MCDBR	28.0	31.9	32.3	30.4	28.5	25.7	25.0	25.3	26.2	30.5	29.8	27.5
		MCWBR	18.0	19.4	17.3	15.4	11.0	9.4	9.0	10.0	9.8	15.0	16.8	18.4
	5% WB	MCDBR	25.7	29.2	26.4	26.1	23.5	22.3	21.9	21.5	21.9	23.7	27.1	25.7
		MCWBR	17.1	18.5	15.5	14.8	11.1	10.6	9.9	9.6	9.9	13.9	17.0	18.0
Clear-Sky Solar Irradiance	taub	0.281	0.289	0.320	0.359	0.386	0.395	0.414	0.413	0.374	0.332	0.287	0.276	
	taud	2.482	2.470	2.421	2.338	2.336	2.360	2.317	2.318	2.376	2.477	2.525	2.494	
	Ebn at Noon	284	297	297	290	283	279	273	270	274	277	279	277	
	Edh at Noon	24	28	33	39	40	39	40	39	35	28	23	22	
All-Sky Solar Radiation	RadAvg	714	975	1305	1639	1904	2156	2124	1868	1525	1101	793	606	
	RadStd	62	76	102	96	122	116	123	97	95	111	58	59	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page